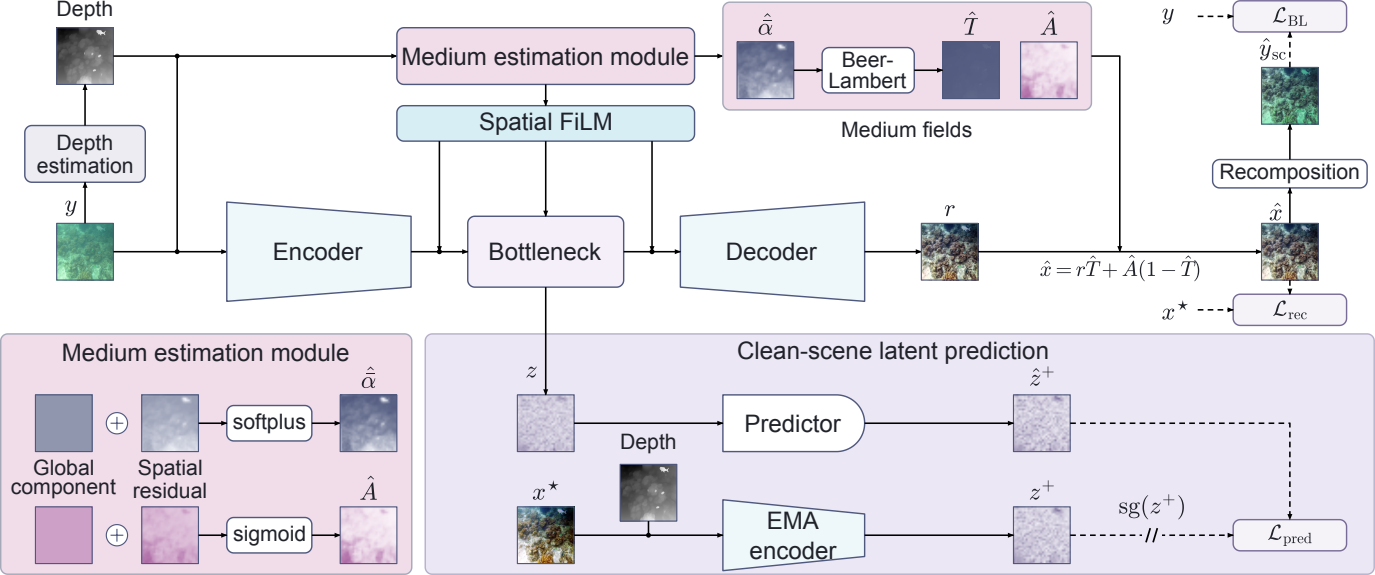

Depth
Depth estimation
y
Medium estimation module
Spatial FiLM
Encoder
Bottleneck
Decoder
$\hat{\tilde{\alpha}}$
$\hat{T}$
$\hat{A}$
Beer-Lambert
Medium fields
r
$\hat{x} = r\hat{T} + \hat{A}(1-\hat{T})$
y
$\mathcal{L}_{\text{BL}}$
$\hat{y}_{\text{sc}}$
Recomposition
$\hat{x}$
$x^\star$
$\mathcal{L}_{\text{rec}}$
Medium estimation module
Global component
Spatial residual
softplus
sigmoid
$\hat{\tilde{\alpha}}$
$\hat{A}$
Clean-scene latent prediction
z
Predictor
$\hat{z}^+$
Depth
$x^\star$
EMA encoder
z^+
$\text{sg}(z^+)$
$\mathcal{L}_{\text{pred}}$